

Computer Graphics:

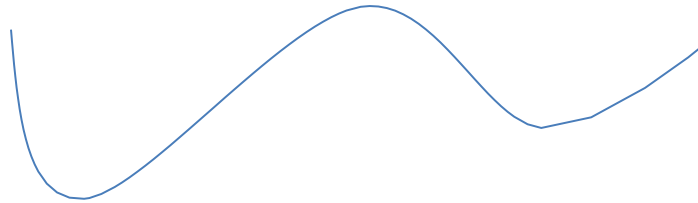
Curves

Topics

- What is a curve and its type?
- Parametric Curve
- Parametric & Geometric Continuity
- Spline Curve
- HERMITE Spline Curve
- Bezier Spline Curve
- B-Spline Curve

Curves & its Presentation

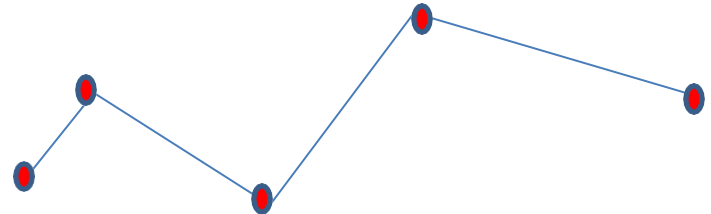
When we think about curve following geometric figure comes into mind



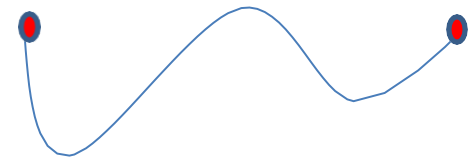
But Curve is more than this!

Definition:

1. Lines joining a set of points cont.



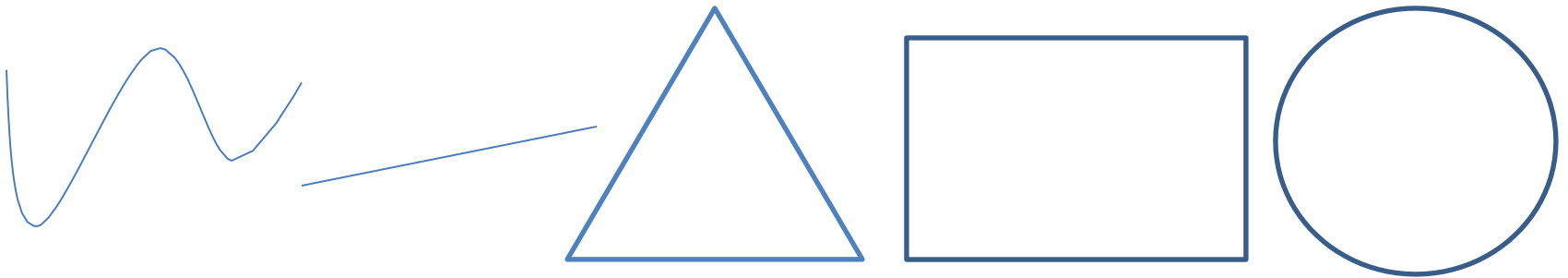
1. Travel one point to another without any gap



Definition: A curve is a set of infinite points in which each point has two neighbours except end points

Curves & its Presentation

Example of curves



Types:

1. **Explicit**

A mathematical function $y = f(x)$ can be plotted as a curve. Such a function is the explicit representation of the curve. The explicit representation is not general, since it cannot represent vertical lines and is also single-valued. For each value of x , only a single value of y is normally computed by the function

2. **Implicit**

Implicit curve representations define the set of points on a curve by employing a procedure that can test to see if a point is on the curve. Usually, an implicit curve is defined by an implicit function of the form, $f(x,y)=0$

A common example is the circle, whose implicit representation is

$$x^2 + y^2 - R^2 = 0$$

Curves & its Presentation

Types:

3. Parametric Curve

Curves having parametric form are called parametric curves. The explicit and implicit curve representations can be used only when the function is known. In practice the parametric curves are used. A two-dimensional parametric curve has the following form –

$$P(t) = f(t), g(t) \text{ or } P(t) = x(t), y(t)$$

The functions f and g become the (x,y) coordinates of any point on the curve, and the points are obtained when the parameter t is varied over a certain interval $[a, b]$, normally $[0, 1]$.

$$P(t=0.8)$$

$$(x_0, y_0), (x_1, y_1)$$

$$P(x,y) = (1-t)P_0 + tP_1$$

$$x = (1-0.8) * x_0 + 0.8 * x_1 = 0.2 * x_0 +$$

$$0.8 * x_1 \quad y = (1-0.8) * y_0 + 0.8 * y_1 = 0.2 * y_0 +$$

$$0.8 * y_1$$

Curves & its Presentation

Types:

3. Parametric Curve

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$$x = (1-0.8) * x_0 + 0.8 * x_1 = 0.2 * x_0 +$$

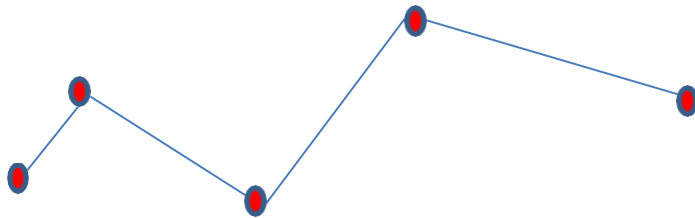
$$0.8 * x_1 \quad y = (1-0.8) * y_0 + 0.8 * y_1 = 0.2 * y_0 +$$

$$0.8 * y_1$$

Design of Curves

Types:

1. **Piece-wise linear:** Join points by line (linearly joined) : ex: $y=mx+c$



but it is not smooth and not realistic.

2. **Piece-wise Polynomial:** Join points with polynomial (non-linear) curve. Ex: $y = x^3 + x^2 + 6$ polynomial with degree=3 needs 4 points to draw. Remember $y=mx+c$ is a polynomial with degree=1 needs 2 points. Use 4-5 points gives the best result. For 4 point curve called cubic

Parametric & Geometric Continuity

Quality of curve means how much is it smooth or continuous

Parametric Continuity: Checks the continuity when join two parametric curve

Last point of P joins the first point of Q

To check it use the following methods

1. Zero Order Parametric Continuity (C^0):

Satisfy the following conditions

$$P(1) = Q(0)$$

Last point of P and 1st point of Q must be same.

2. First Order Parametric Continuity (C^1):

Satisfy the following conditions

$$P'(1) = Q'(0)$$

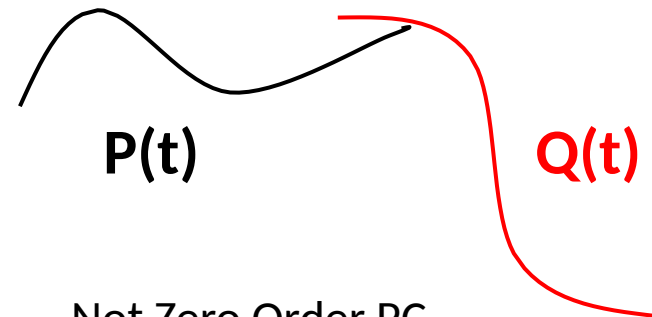
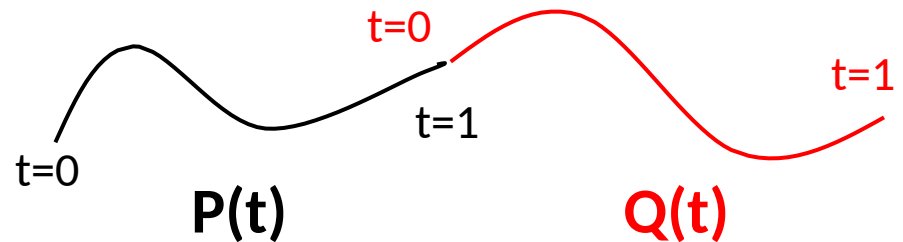
The tangent at Last point of P and 1st point of Q must be same.

3. 2nd Order Parametric Continuity (C^2):

Satisfy the following conditions

$$P''(1) = Q''(0) \text{ and } P'(1) = Q'(0)$$

For C^2 it must satisfy C^0 and C^1 .



Not Zero Order PC
 $P(1) \neq Q(0)$

Parametric & Geometric Continuity

Geometric Continuity: Checks the continuity according to the shape when join two curves

To check it use the following methods

1. Zero Order Geometric Continuity

(G^0): Same as C^0

$$P(1) = Q(0)$$

2. First Order Geometric Continuity

(G^1): Satisfy the following conditions

$$A'(1) \propto B'(0)$$

A: Last point of P and B: 1st point of Q.

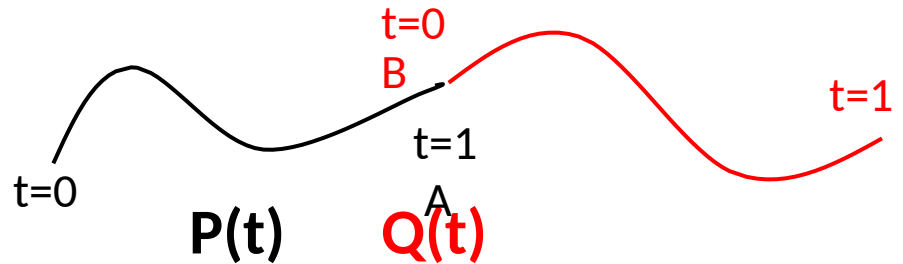
Both tangent must be the same

direction, $C^1 \Rightarrow G^1$ but $G^1 \not\Rightarrow C^1$

3. 2nd Order Parametric Continuity

(G^2): Satisfy the following conditions

$$P''(1) \propto Q''(0)$$



Let $A(x_p, y_p)$ and $B(x_q, y_q)$



Spline Curves

Spline Curve: It is a strip of Balsha tree used to design a ship using ducks. It is a type of curve in which control points control the shape of the curve.

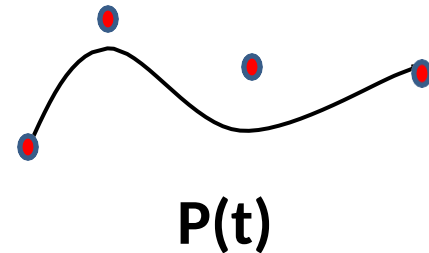
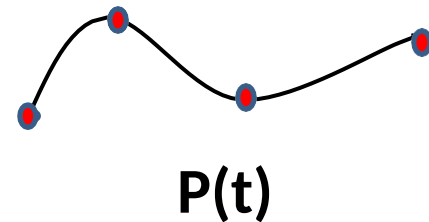


Types

1. Interpolation Spline Curve: It passes through the control points. Example of this type of curve is HERMIT curve

2. Approximation Spline Curve

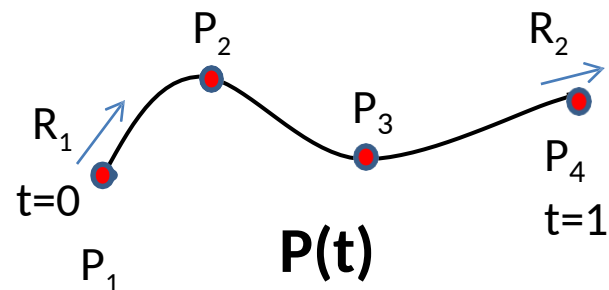
A curve that passes near to the control points but not necessarily passes through these points. Example Biezer Curve.



Hermite Spline Curves

Hermite Spline Curve:

- It is Interpolation spline curve
- It uses cubic polynomial function (uses 4 points)
- It uses two position vector (end-points), P1 & P2
- It uses two tangent vector (in which position curve is moving) at end points, R1 & R2



Let $Q(t)$ is the curve at point $[x(t), y(t), z(t)]$

Then we can write in matrix form

$$Q(t) = [x(t) \quad y(t) \quad z(t)]$$

$$x(t) = a_x t^3 + b_x t^2 + c_x t + d_x$$

$$y(t) = a_y t^3 + b_y t^2 + c_y t + d_y \quad \text{where } R_1(t=0)=P_1' \text{ and } R_2(t=1)=P_2'$$

$$z(t) = a_z t^3 + b_z t^2 + c_z t + d_z$$

$$T = [t^3 \quad t^2 \quad t \quad 1] \quad C = \begin{bmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \\ d_x & d_y & d_z \end{bmatrix}$$

$$Q(t) = T \cdot C$$

$$Q(t) = T \cdot C$$

Here $C = M \cdot G$

where M : Basis M (confirms C^1 & C^2 continuity)

G : Geometric Vector Matrix

$$C = T \cdot M_H \cdot G_H \quad R_1(t=0)=P_1'$$

$$Q_x(t) = P_1(t=0) = [0 \quad 0 \quad 0 \quad 1] \cdot M_H \cdot G_H$$

$$Q_x(t) = P_1(t=1) = [1 \quad 1 \quad 1 \quad 1] \cdot M_H \cdot G_H$$

$$Q'_x(t) = R_1(t=0) = [0 \quad 0 \quad 1 \quad 0] \cdot M_H \cdot G_H$$

$$Q'_x(t) = R_1(t=1) = [3 \quad 2 \quad 1 \quad 0] \cdot M_H \cdot G_H$$

Hermite Spline Curves

$$T = \begin{bmatrix} t^3 & t^2 & t & 1 \end{bmatrix} \quad C = \begin{bmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \\ d_x & d_y & d_z \end{bmatrix}$$

$$C = T \cdot M_H \cdot G_H$$

$$Q_x(t) = P_1(t=0) = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \cdot M_H \cdot G_H$$

$$Q_x(t) = P_1(t=1) = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \cdot M_H \cdot G_H$$

$$Q'_x(t) = R_1(t=0) = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \cdot M_H \cdot G_H$$

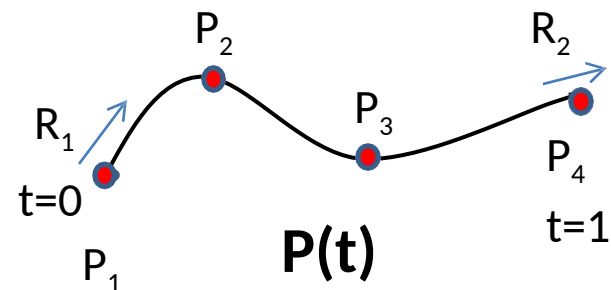
$$Q'_x(t) = R_1(t=1) = \begin{bmatrix} 3 & 2 & 1 & 0 \end{bmatrix} \cdot M_H \cdot G_H$$

The above four equations can be written as

$$\begin{bmatrix} P_1 \\ P_4 \\ R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix} \cdot M_H \cdot G_H$$

$$M_H \cdot G_H = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix}^{-1} \cdot \begin{bmatrix} P_1 \\ P_4 \\ R_1 \\ R_2 \end{bmatrix}$$

$$M_H = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 3 & 2 & 1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 2 & -2 & 1 & 1 \\ -3 & 3 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad G_H = \begin{bmatrix} P_1 \\ P_4 \\ R_1 \\ R_2 \end{bmatrix}$$



$$Q(t) = \begin{bmatrix} t^3 & t^2 & t & 1 \end{bmatrix} \cdot \begin{bmatrix} 2 & -2 & 1 & 1 \\ -3 & 3 & -2 & -1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} P_1 \\ P_4 \\ R_1 \\ R_2 \end{bmatrix}$$

$$= (t^3 - 3t^2 - 1)P_1 + (-2t^3 + 3t^2)P_4 + (t^3 - 2t^2 + t)R_1 + (t^3 - t^2)R_2$$

$$Q(t) = H_1 P_1 + H_2 P_4 + H_3 R_1 + H_4 R_2$$

$$H = \begin{bmatrix} H_1 & H_2 & H_3 & H_4 \end{bmatrix}$$

H=Hermite Blending

Hermite Curve: Example

★ A cubic splined curve has start point at $P_0(16, 0)$ and end point at $P_1(3, 1)$. The tangent vector for end point P_0 is given by line joining and point $P_2(14, 8)$. Tangent vector is given by line joining P_2 and point P_1 .

1. Determine the parametric equation of HCC
2. Plot the HCC

⇒ Equation for Hermite Cubic Spline Curves:

$$P(u) = [P_0 \ P_1 \ P_0' \ P_1'] \begin{bmatrix} 2 & -3 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u^3 \\ u^2 \\ u \\ 1 \end{bmatrix}$$

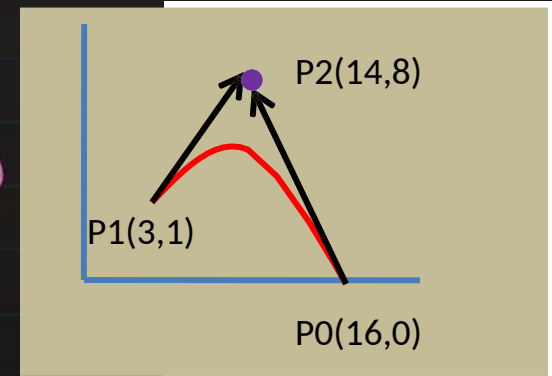
P_0 = Starting point = $(16, 0)$

P_1 = end point = $(3, 1)$

P_2 = Intersecting point of tangents = $(14, 8)$

P_0' = Tangent at start point

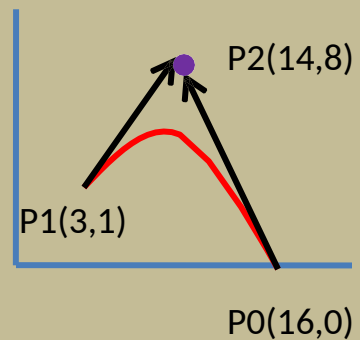
P_1' = Tangent at end point



Hermite Curve: Example contd

$$\text{So, } P'_0 = P_2 - P_0 = \begin{bmatrix} 14 & 8 \end{bmatrix} - \begin{bmatrix} 16 & 0 \end{bmatrix} \\ = \begin{bmatrix} -2 & 8 \end{bmatrix}$$

$$P'_1 = P_1 - P_2 = \begin{bmatrix} 3 & 1 \end{bmatrix} - \begin{bmatrix} 14 & 8 \end{bmatrix} \\ = \begin{bmatrix} -11 & 7 \end{bmatrix}$$



$$P(u) = [P_0 \ P_1 \ P'_0 \ P'_1] \begin{bmatrix} 2 & -3 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u^3 \\ u^2 \\ u \\ 1 \end{bmatrix}$$

for X coordinates of curves

$$P_x(u) = \begin{bmatrix} 16 & 3 & -2 & -11 \end{bmatrix} \begin{bmatrix} 2 & -3 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u^3 \\ u^2 \\ u \\ 1 \end{bmatrix}$$

$P_0 = (16, 0)$
 $P_1 = (3, 1)$
 $P'_0 = (16, 0)$
 $P'_1 = (-11, 7)$

$$P_x(u) = 13u^3 - 24u^2 - 2u + 16$$

Hermite Curve: Example contd

for y coordinates

$$P_y(u) = [0 \ 1 \ 8 \ -7] \begin{bmatrix} 2 & -3 & 0 & 1 \\ -2 & 3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u^3 \\ u^2 \\ u \\ 1 \end{bmatrix}$$

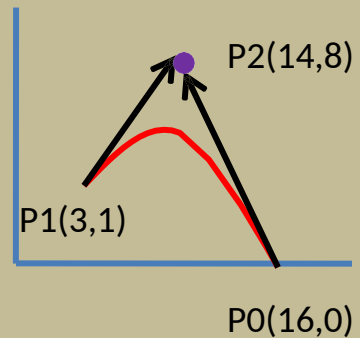
$$P_y(u) = [-1 \ -6 \ 8 \ 0] \begin{bmatrix} u^3 \\ u^2 \\ u \\ 1 \end{bmatrix}$$

$$\therefore P_y(u) = -u^3 - 6u^2 + 8u$$

Hence the parametric equations are:-

$$P_x(u) = 13u^3 - 24u^2 - 2u + 16$$

$$P_y(u) = -u^3 - 6u^2 + 8u$$



$$P_0 = (16, 0)$$

$$P_1 = (3, 1)$$

$$P'_0 = (16, 0)$$

$$P'_1 = (-11, 7)$$

Bezier Curves

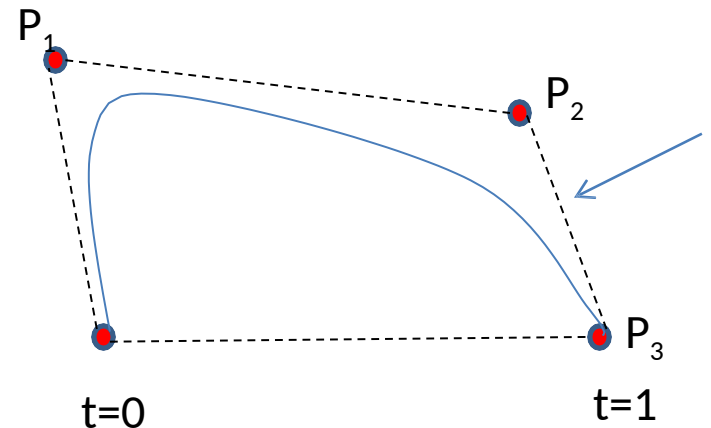
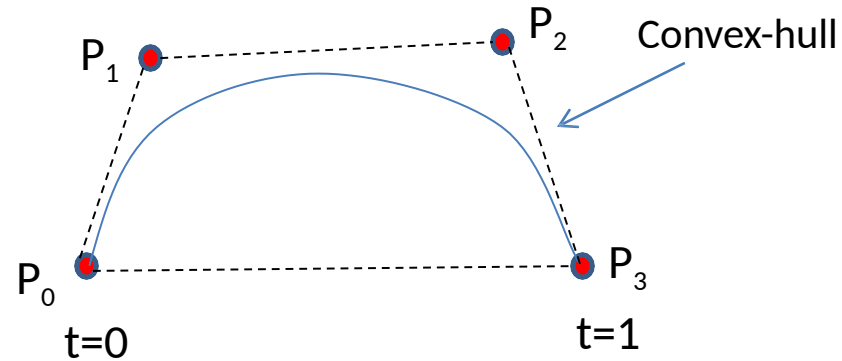
Properties of Bezier Curve

1. It passes first and last points
2. The curve must be contained in the convex hull (connecting control points) of the control points
3. The control points control the shape
4. The degree of polynomial is one less than Control points.
5. It exhibits global control means if any control points moves then whole shape of the curve changed.

If control points=2 then degree=1 then polynomial is linear

If two control points=3 then degree=2 then polynomial is quadratic

If two control points=4 then degree=3 then polynomial is cubic



Bezier Curves

Let consider 2 control points P_0 and P_1 then
Any point L_0 between P_1 and P_2

$$L_0(t) = (1 - t)P_0 + tP_1$$

Consider 3 control points P_0 and P_1 and P_2

$$L_1(t) = (1 - t)P_1 + tP_2$$

At $t=x$, calculated point at P_0P_1 line is L_0 , at P_1P_2 line is L_1 and at point AB line is Q_0 . The points L_0 , L_1 and Q_0 change with t . The final curve follows the point Q_0 .

$$Q_0(t) = (1 - t)L_0(t) + tL_1(t)$$

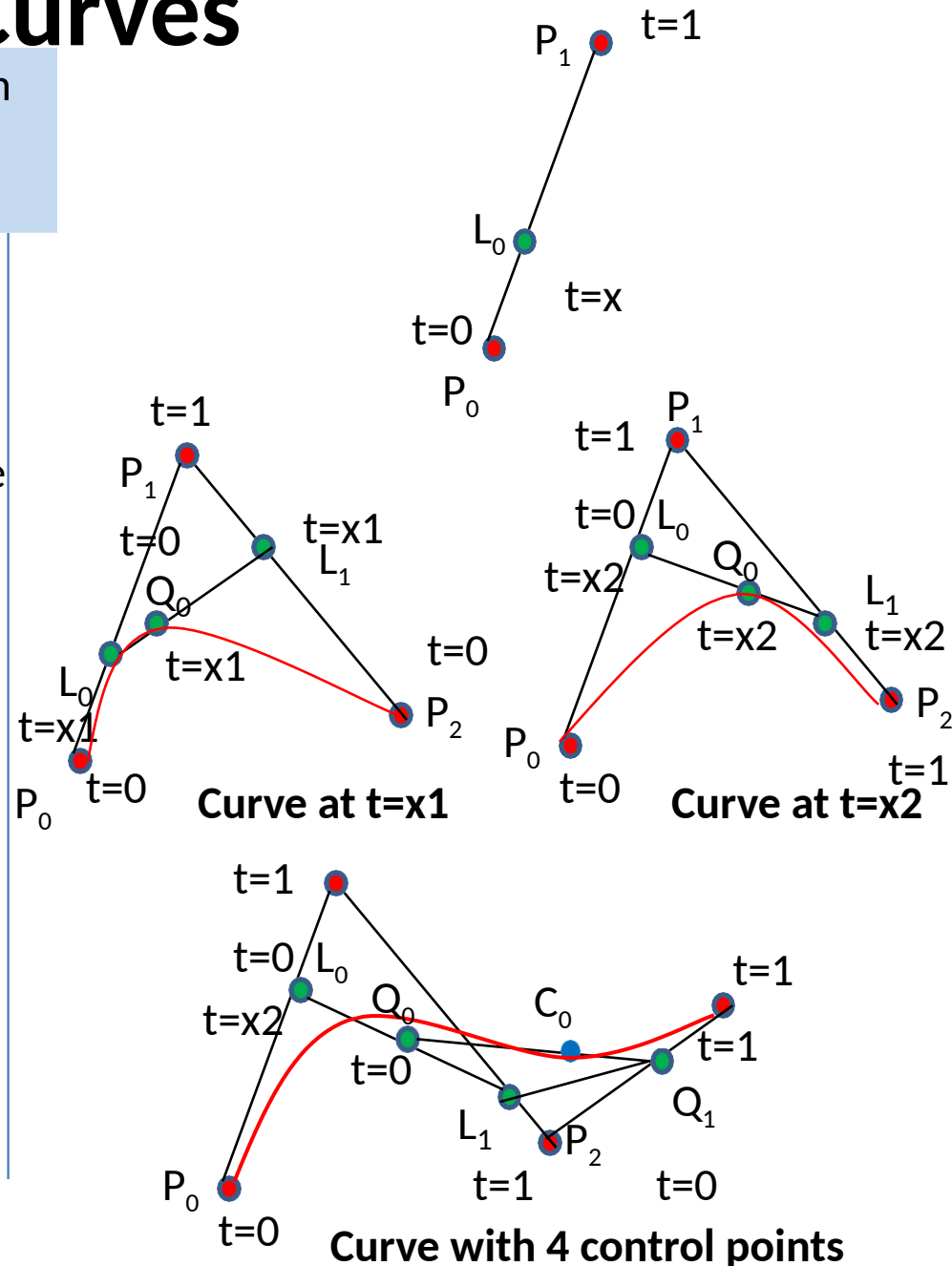
$$Q_0(t) = (1 - t)[(1 - t)P_0 + tP_1] + t[(1 - t)P_1 + tP_2]$$

$$Q_0(t) = ((1 - t)^2 P_0 + 2(1 - t)tP_1 + t^2 P_2)$$

For 4 control points P_0, P_1, P_2 and P_3

$$C_0(t) = (1 - t)Q_0(t) + tQ_1(t)$$

$$C_0(t) = ((1 - t)^3 P_0 + 3(1 - t)^2 tP_1 + 3(1 - t)t^2 P_2 + t^3 P_3)$$



Prob: Design a Bezier curve controlled by 4 points $P_0(1,1)$, $P_1(2,3)$, $P_2(4,3)$ and $P_3(6,4)$.

Soln:

For 4 Control Points (CP) P_0, P_1, P_2 and P_3

Degree of polynomial, $n=4-1=3$

The equation of Bezier Curve for 4 CPs is

$$C_0(t) = ((1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t) t^2 P_2 + t^3 P_3$$

Solving for x $x_0 = 1, x_1 = 2, x_2 = 4, x_3 = 6$

$$P_x(u) = x_0(1-u)^3 + 3x_1u(1-u)^2 + 3x_2u^2(1-u) + x_3u^3$$

$$= (1-u)^3 + 6u(1-u)^2 + 12u^2(1-u) + 6u^3$$

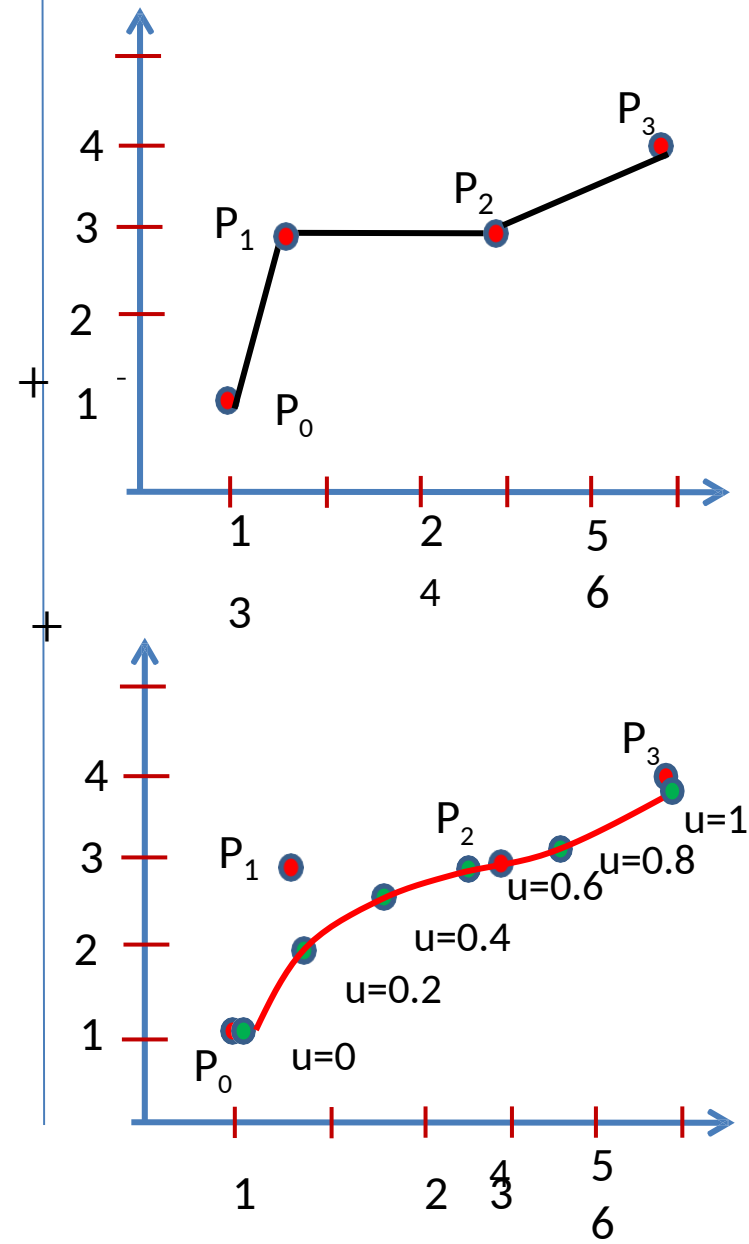
Solving for y $y_0 = 1, y_1 = 3, y_2 = 3, y_3 = 4$

$$P_y(u) = y_0(1-u)^3 + 3y_1u(1-u)^2 + 3y_2u^2(1-u) + y_3u^3$$

$$= (1-u)^3 + 9u(1-u)^2 + 9u^2(1-u) + 4u^3$$

u	$P_x(u)$	$P_y(u)$
0.0	1	1
0.2	1.712	1.984
0.4	2.616	2.632
0.6	3.664	3.088
0.8	4.808	3.496
1.0	6	4

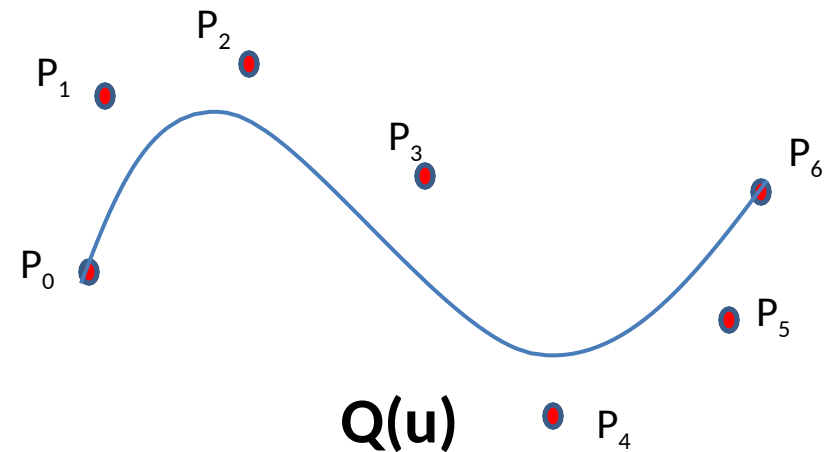
Bezier Curves



B-Spline Curves

Properties of B Spline Curve

- It is an approximate Spline curve
- It's local control curve by control points
- The curve is divided into segments and any of control point (CP) affects only in the segment. The number of segment is obtained by
- $\text{Seg} = n - k + 2$ here $n = \text{total CP} - 1$, $k = \text{no. of control points in a segment}$ ($2 \leq k \leq \text{CP}$) Let $k=3$ then $\text{seg} = (7-1)-3+2=5$



Segment Control Points

Q_1 $P_0 P_1 P_2$

Q_2 $P_1 P_2 P_3$

Q_3 $P_2 P_3 P_4$

Q_4 $P_3 P_4 P_5$

Q_5 $P_4 P_5 P_6$

- Degree of the polynomial $= k - 1 = 3 - 1 = 2$ The curve will be piecewise polynomial of degree $(k-1)$

B-Spline Curves

General equation of B Spline Curve

$$Q(u) = \sum_{i=0}^n P_i * N_{i,k}(u)$$

Where

P_i = control points

u = parameter ($0 \leq u \leq n-k+2$)

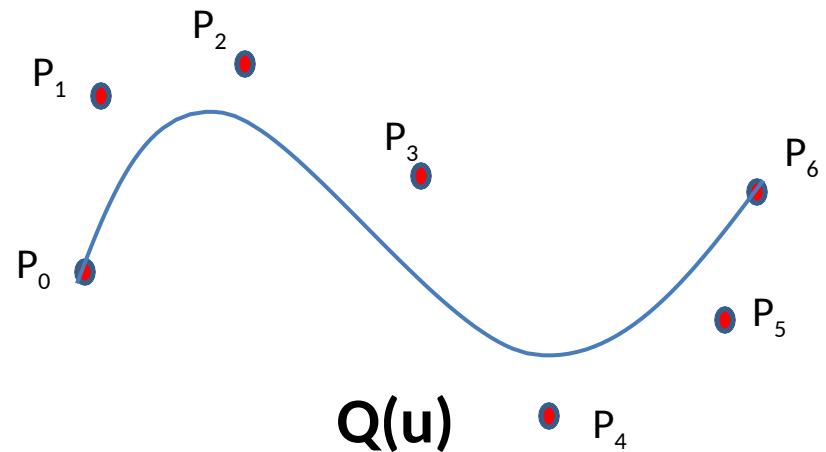
$Q(u)$ has two values $x(u)$ and $y(u)$, then

we can write

$$x(u) = \sum_{i=0}^n x_i * N_{i,k}(u) \quad y(u) = \sum_{i=0}^n y_i * N_{i,k}(u)$$

$$N_{i,k}(u) = \frac{(u - t_i) N_{i,k-1}(u)}{t_{i+k-1} - t_i} + \frac{(t_{i+k} - u) N_{i+1,k-1}(u)}{t_{i+k} - t_{i+1}}$$

$$N_{i,k}(u) = \begin{cases} 1, & t_i \leq u \leq t_{i+1} \\ 0, & \text{otherwise} \end{cases}$$



t_i = Joint Point at which piece of curve joins

$i = (0 \leq i \leq$

$n+k)$ $t_i = 0$ if $i < k$

$t_i = i - k + 1$ if $k < i < n$

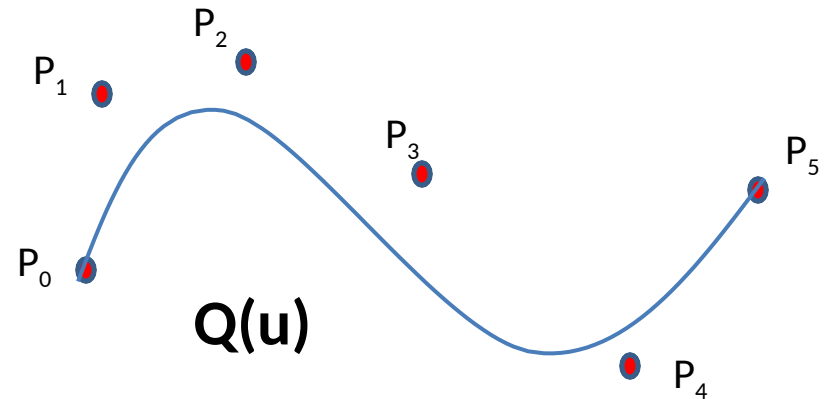
$t_i = n - k + 2$ if $i > k$

B-Spline Curves

Example : Here CP=6, n=5 and k=3, the equation of the curve

$$Q(u) = \sum_{i=0}^5 P_i * N_{i,k}(u)$$

$$= P_0 N_{0,3}(u) + P_1 N_{1,3}(u) + P_2 N_{2,3}(u) + P_3 N_{3,3}(u) + P_4 N_{4,3}(u) + P_5 N_{5,3}(u)$$



Here $u=n-k+2=4$, so range of $u=(0 \leq u \leq$

$$Q_1(u) = P_0 N_{0,3}(u) + P_1 N_{1,3}(u) + P_2 N_{2,3}(u) \quad 0 \leq u \leq 1$$

$$Q_2(u) = P_1 N_{1,3}(u) + P_2 N_{2,3}(u) + P_3 N_{3,3}(u) \quad 1 \leq u \leq 2$$

$$Q_3(u) = P_2 N_{2,3}(u) + P_3 N_{3,3}(u) + P_4 N_{4,3}(u) \quad 2 \leq u \leq 3$$

$$Q_4(u) = P_3 N_{3,3}(u) + P_4 N_{4,3}(u) + P_5 N_{5,3}(u) \quad 3 \leq u \leq 4$$

t_i = Joint Point at which piece of curve joins

$i = (0 \leq i \leq n+k=8)$

Then we need to find $[t_0, t_1, \dots, t_8]$

$t_i = 0$ if $i < k$

$t_i = i - k + 1$ if $k < i < n$

$t_i = n - k + 2$ if $i > k$

$t_0 = 0$ as $0 < 3$ ($i < k$)

$t_1 = 0$ as $1 < 3$ ($i < k$)

$t_2 = 0$ as $2 < 3$ ($i < k$)

$t_3 = 1$ as $3 \leq 3 < 5$ ($k \leq i < n$) $t_4 = 4$

as $4 > 3$ ($i > k$)

$t_5 = 4$ as $5 > 3$ ($i > k$)

$t_6 = 4$ as $6 > 3$ ($i > k$)